

HoloType: Lived Experience Based Communication Training for Nonspeaking Autistic People

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ABSTRACT

The lived experiences of nonspeaking autistic people are typically not reflected in technology designed to support them. In this paper, we describe insights from nonspeaking autistic people who have learned to communicate by typing and those who support them. We leverage these within an augmented reality communication training prototype. The prototype provides an engaging, personalized learning environment in which a user can systematically develop their pointing skills. Pointing skills combined with elements of language such as spelling can in turn facilitate expressive written communication skills. We describe our requirements elicitation process and provide details about our initial implementation.

CCS CONCEPTS

• **Software and its engineering** → **Software creation and management**; • **Designing software** → *Requirements analysis*.

KEYWORDS

assistive communication technology, augmented reality, autism

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1 INTRODUCTION

Autism is a lifelong, neurological condition that affects different people in very different ways. Although it is most commonly associated with social differences, difficulties with communication are also common. Indeed, about one-third of autistic children and adults are “nonspeaking”: They cannot communicate effectively

using speech [7]. Some individuals can say a few words or phrases, or repeat back material they have heard before, but many do not communicate using speech at all. We do not yet know why some autistic people cannot communicate effectively using speech. But given the well-documented motor challenges many autistic people face [8, 9, 29] and given that speech requires precise motor coordination of the vocal apparatus, one promising hypothesis is that difficulties stem from differences in nonspeaking autistic people’s motor system [26].

Unfortunately, most nonspeaking autistic people are never provided access to an effective, language-based alternative method of communication [4]. As a result, they are deprived of educational, social, employment, and community opportunities that most people who can speak take for granted. Without access to a conventional language-based method of communication, their ability to participate in medical decision-making or planning for their future is also severely limited.

Some nonspeaking autistic people have gained access to an effective alternative to speech: These individuals have learned, over the course of many years, to communicate by typing or pointing to letters. Some people who communicate in these ways have graduated from college [24]; others have written best-selling memoirs [10]. Their insights into autism and communication have been remarkable. As one example, their written testimony supports the motor hypothesis described earlier: Many have described a disconnect between what they would like their bodies to do and what their bodies actually do [10, 11, 27], and they have reported on the extraordinary effort it takes for them to produce intelligible speech [9].

No existing computer-based training system embodies such insights gleaned from the lived experience of nonspeaking autistic people, their families, therapists, and educators. In this paper, we present a prototype system called *HoloType* that addresses this vacuum. Our key idea is to exploit Augmented/Mixed Reality (AR/MR) technology in combination with wearable devices to realize a communication training system for nonspeaking autistic people. The system engages a user in interesting, interactive, and age-appropriate educational content. Interaction with the content is tailored based on the motor skills of the user. Furthermore, the content is structured to systematically improve motor skills, i.e., ability to point to virtual objects, while simultaneously focusing on other aspects such as spelling, math and science. Eye, hand, and

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head tracking data from the AR device in combination with data from wearable sensors are used to assess the skills learnt and adapt the content accordingly as well as to provide attentional and sensory regulation supports. We describe our requirements elicitation process and detail the features we have already implemented.

2 BACKGROUND AND RELATED WORK

2.1 Testimonials from nonspeaking autistic people

We do not yet know why some autistic people are nonspeaking. However, nonspeaking autistic people who type or point to letters have provided valuable insights into the kinds of training and practices that they found helpful as they learned to communicate in these ways. It is these insights that have provided the foundation for the HoloType prototype. We described four of them here:

Presuming competence: Educational research with non-autistic children makes clear the importance of setting and maintaining high expectations [22]: Presuming that students can learn when provided appropriate kinds of support impacts both the kinds of learning opportunities students are offered and the implicit and explicit messaging they receive about their potential. Unfortunately, many nonspeaking autistic individuals have reported that many of the educators with whom they worked did not presume competence—did not presume that they could learn with the right kinds of support. For example, one nonspeaking autistic person explained that before he learned to communicate by pointing to letters, his education consisted largely of drills: “If I had a dollar for every time I had to touch my nose, I’d be rich. I remember one day [the teachers] realized that I hated being told to touch my nose, so they brilliantly switched the command to ‘touch your head!’ I felt like a prisoner of these theories and methods in sessions lasting eight hours a day, forty hours a week, year in and year out” [11]. Rather than being repetitively drilled on meaningless tasks, nonspeaking autistic people have urged that students be offered support to learn age-appropriate content, which they unsurprisingly report is more interesting and motivating.

Open-ended communication: Some augmentative and alternative communication (AAC) systems provided to nonspeaking people make use of pictures or icons. For example, the early training phases of a commonly used AAC system called the Picture Exchange Communication System (PECS) require a user to exchange a picture of e.g., a cookie for an actual cookie [3]. PECS and systems like it can be useful for making requests, but they are limited in at least two ways. First, the vocabulary available is chosen by someone else for the user and so these icon- or picture-based systems do not allow open-ended communication. Second, although later training phases of PECS are intended to teach users to ask questions and make comments (using the available vocabulary), few PECS users progress beyond the requesting phase [25]. Nonspeaking people who have learned to type or point to letters to communicate have emphasized the importance of learning to use a robust AAC system—namely, spelling—that allows them to say whatever they want [17].

Practicing purposeful motor control: As noted earlier, a good deal of research has shown that autistic people have significant

motor challenges, which may be one reason why some cannot communicate effectively using speech. Typing or pointing to letters also requires motor control—specifically, the ability to accurately and precisely isolate the target letter amidst 25 non-target letters and to generate and execute the movement of the arm to that letter. Interestingly, most autistic children do not point, even to request desired objects or toys [19]. One nonspeaking autistic individual who learned to communicate by pointing to letters, handwriting, and typing explained that as a child, he “could not point at objects for many reasons. The most important reason is that I had very little sensation of my body” [1]. But it is a skill that can be taught. Nonspeaking people who have learned to type or point to letters have explained the importance of having regular, individualized, and systematic opportunities to practice pointing skills—situated in meaningful age-appropriate content [11].

Attentional and sensory regulation strategies: In addition to communication and motor challenges, many autistic people have difficulties with attention, impulsivity, and restlessness; indeed, ADHD is a commonly co-occurring condition, affecting as many as 50% of autistic children [14]. Differences in sensory systems are also common, affecting as many as 90% of autistic individuals [19]. In some cases, autistic people can be overwhelmed by sensory stimuli, including lights and sounds [20]. Some nonspeaking autistic people have described the kinds of strategies they use to mitigate these attentional and sensory challenges. For example, they report being able to attend more effectively when they are allowed to move around than when they are not allowed to do so [10]. (For empirical research confirming attentional benefits of higher levels of activity in students with ADHD, see [23]). They also report that interacting with (and sometimes the mere presence of) a trusted person confers a multitude of regulatory benefits, including allowing them to participate in environments they would otherwise find overwhelming [11, 17]. (For a review of empirical research confirming the regulatory benefits of being with a trusted other—particularly in potentially stressful situations, see [6]).

2.2 Existing approaches

Traditional communication approaches do not account for the significant sensory-motor-attentional challenges that many autistic people face [8, 9, 12, 13, 20]. For the nonspeaking autistic population, the most common communication interventions involve training on the use of speech or picture-based systems [5], approaches that have not been shown to be effective for most nonspeaking people. A paper reviewed the only two randomized controlled trials (the gold standard of intervention research) involving these methods [4]. The conclusion was that neither intervention “*resulted in improvements in verbal or non-verbal communication that were maintained over time*” [4].

Many recent studies have explored the use of virtual, mixed, and augmented reality technology in the autism space. But most of these approaches target only social, emotional, and daily life skills [15]; they do not address the urgent need for providing nonspeaking people with access to an expressive, language-based means of communication. Contrary to consistent testimonials from this population, none of today’s solutions focus on motor skills training *situated with meaningful, age-appropriate lessons that teach spelling skills to foster open-ended communication.*

3 REQUIREMENTS ELICITATION

We consulted with experts to obtain strategies to translate the high-level insights gleaned from nonspeaking autistic people discussed in Section 1 to a prototype training system. Our group included an occupational therapist who had 25 years of experience teaching non-speaking autistic people to communicate by building their pointing skills. It also consisted of two parents who had coached their non-speaking autistic children to type. We next outline the key requirements that emerged out of these sessions.

3.1 Requirements

R1: Age-appropriate and customizable lessons – The experts corroborated non-speaker testimonials about the need to situate motor skills and spelling training within an engaging, age-appropriate, individualized lesson. A lesson chosen in this manner piques the interests of the learner leading to better engagement.

As an example, an educator spoke about their initial sessions with a 20-year-old minimally verbal student. That student was never placed in age-appropriate classrooms. They were placed in special education classrooms where the emphasis was more on life skills training. Furthermore, the student had only basic motor skills as they pertain to pointing. During the first meeting with this student's parents, the educator learnt that the student enjoys listening to the music group Boney M. This motivated the educator to prepare a lesson based on the group. Excerpts from this lesson are shown in Table 1.

The design of this lesson illustrates several key requirements. First, lessons must be individualized by an experienced educator. In this case, content about Boney M played a significant part in the student being engaged with the session. Second, the intrinsic reward of an interesting lesson is leveraged by the educator to teach motor skills. Specifically, due to the still evolving pointing skills of the student, they are required to interact with the lesson only through 2 choices. For example, the lesson trains the student to move one of their fingers to choose between one of two choices ("main" or "extra"; "play the drums" or "compose music"). Initially, the educator might model the action of choosing the correct response either by pointing at the correct choice themselves or moving the finger of the student to the correct choice. These prompts and modeling actions are gradually faded away as the student improves their pointing skills. Third, the lesson is also used by the educator to teach spelling, a key foundation of expressive written language. For example, in the lesson shown in Table 1 the educator spells out the words "Germany" and "Song". Finally, the educator leverages the lesson to teach concepts related to geography (locating Caribbean islands and Germany on a map) and Math ("how many years ago was Boney M formed?").

It is worth noting that the lesson embodies the notion of presuming competence discussed in Section 1. Even though the student was never exposed to regular classrooms, the lesson does not make the unfounded assumption that they are incapable of learning interesting content. It presumes that the student can learn spelling, geography, and math if those concepts are taught within an engaging framework adapted to their motor skills. Furthermore, it considers the age and interests of the student. For example, even though the student was never exposed to spelling in a significant

way at school, the lesson eschews infantilizing them by asking them to spell words that are more appropriate for younger learners (e.g., "cat" or "dog").

R2: Customization for motor skills – Our consultations revealed an overarching graduated strategy for building pointing skills. The experts reported success in taking a strength-based approach that adapts the way a learner interacts with a lesson to their current motor skills level. This contrasts with an approach where the learner is prevented from accessing engaging educational content in a meaningful way due to their as-yet underdeveloped motor skills. All experts spoke about a thoughtful progression of lesson interactions that promotes learning of complex concepts, language, and spelling whilst simultaneously improving pointing skills. Specifically, we were able to distill three different interaction modes described as follows:

Beginner - As in the Boney M lesson segment, learners interact with the lesson using 2 choices. At this level, visual clutter is reduced by offering the learner only 2 choices for every question. Educators implement choices using large objects, e.g., very large pieces of paper with the choices written on them or large drawings on an easel board. They also emphasized the need for generous spacing between the choice objects. These accommodations account for learners who may not yet possess evolved gross motor skills. As discussed previously, the educator would initially provide prompts or model the act of pointing to the correct choices. The learner interacts via choices while answering questions related to spelling and those related to the lesson. For example, consider the interaction "Let's spell the word Boney" from Table 1. The educator would try to elicit the correct spelling by offering a sequence of choices ("What is the first letter? N or B?" and so on).

Intermediate - In this mode, the number of choices is increased to reflect mastery over basic motor skills demonstrated in the beginner mode. For example, some educators choose to offer 3 choices using slightly smaller objects with lesser spacing between them. Others might offer limited number of choices based on the context. For example, when a learner is asked to spell "Boney", only those letters present in that word are laid out in front of them in a random order. The learner is then asked to point to the choices to spell out the word. The level of prompting offered by the educator is also faded out in this mode.

Advanced - This mode allows learners to express more open-ended responses. Typically, a full letter board or a number board is offered, based on the context. The characters in these devices are significantly smaller than those in the intermediate mode and they are also very close to one another. A learner uses these devices to spell out words and offer descriptive answers to questions. Mastery of this mode allows learners to transition to even more complex devices, e.g., a full functional virtual tablet keyboard.

R3: Customized Regulation Strategies – Several strategies emerged for supporting the sensory and attentional regulation of nonspeaking autistic learners. We describe the main ones as follows:

Simultaneously engaging multiple learning channels – From their practices, the experts pointed us to the importance of engaging a learner using multiple channels. For example, they reported better attention when the educator complements spelling aloud a word with writing out the spelling on a piece of paper displayed in the learner's visual field.

Table 1: Sample Lesson

Educator narration: Boney M was a Euro-Caribbean vocal group created by German record producer Frank Farian, who was the group's primary songwriter.
Educator driven activity: <i>Let's look at where Europe is and where the Caribbean is on the map.</i> <i>Let's spell Germany</i> <i>What is another word that means primary? main or extra?</i>
Educator narration: A songwriter is a musician who professionally composes musical compositions and writes lyrics for songs.
Educator driven activity: <i>So, what is a songwriter likely to do? Play the drums or compose music?</i> <i>Let's spell song.</i>
Educator narration: The group was formed in 1976 and achieved popularity during the disco era of the late 1970s. Since the 1980s, various line-ups of the band have performed with different personnel.
Educator driven activity: <i>If I wanted to see how many years ago Boney M was formed, would I add 1976 to 2021 or would I subtract 1976 from 2021?</i> <i>What is the answer likely to be? 85 or 45?</i>

Customizing the educator's voice – Our experts agreed on the need to individualize the tone, speed, and cadence of the lesson's narration. For example, some learners find a fast narration helpful to overcome attentional issues. However, others might become anxious with such a pace. Thus, the system should allow the flexibility of offering control over voice characteristics.

Adapting for movement – As mentioned in Section 1, many autistic learners move constantly to cope with issues such as sensory overload. Thus, the educator must move with them so that they and the artifacts they present, e.g., choices and spelt out words, remain in the visual performance field of the learner. They also spoke about the need to customize the orientation of input devices, e.g., distance and angle of a letter board, to individual learners to match their movement patterns.

Detecting and mitigating anxiety and attentional issues - Over half of autistic children and adults are thought to have a co-occurring anxiety disorder [21]. This motivates the need for strategies to detect and mitigate anxiety. Confirming nonspeaker testimonials, all experts agreed that interacting with a familiar, trusted educator is important to manage anxiety. Furthermore, the occupational therapist suggested the use of sensory breaks at the first onset of dysregulation as a calming mechanism. For example, a therapist may interrupt the lesson and play a favourite video. To address attentional issues, the experts described the importance of uncluttering the visual field of the learner, e.g., blocking off irrelevant choices while responding to a question. The success of a software-based training system thus depends on its ability to offer customized strategies for regulating anxiety and attentional issues.

R4: Techniques to assess progress– The experts described challenges related to measuring motor skills improvement in a fine-grained manner. They often rely on intuition to track progress and customize the delivery of lessons. However, the lack of objective evaluation criteria can cause problems, e.g., continuing with training at a level that is either too simple or too advanced or using an uncomfortable (to the learner) orientation of input choices.

Thus, novel ways to measure pointing accuracy and user interaction are required to help with better customization of the learning environment.

3.2 Rationale for a Software-Based System

Our experts agreed that there is no substitute for a human educator. However, they were unanimous that a software-based training system such as HoloType could complement their efforts and help learners work towards independent typing. The experts also indicated that individualizing lessons can take considerable time and hence software to help at least partially automate that process would be helpful.

4 SYSTEM IMPLEMENTATION

4.1 Rationale for selecting AR

We deliberated between tablet-based apps and AR for implementing the system. A key advantage of a tablet-based solution is affordability. However, as we discuss shortly, several key requirements identified in Section 3 are difficult to implement on that platform. Although AR devices are expensive right now, as with other computing technology, costs are likely to decrease in future with wider adoption. We summarize the key reasons for selecting AR as follows:

Motor Training - AR allows for large virtual objects, e.g., buttons, for a learner to engage in a lesson without being constrained by fine motor planning and movement difficulties. Indeed, one can design “room scale” buttons to support the beginner mode (R2). Furthermore, given the large field of view, the size of the virtual buttons and the spacing between them can be easily customized.

AR devices such as Microsoft's HoloLens 2 provide hand, head, and eye tracking data. This can be used to implement the prompting needed in the beginner and intermediate modes (R2) to help with motor planning. For example, assume that a learner is moving their finger towards the correct choice but stops for some reason. The system can offer an auditory prompt encouraging them to continue moving towards the target. Consider another example, where a learner is consistently looking at the correct choices, as indicated

by eye gaze tracking, but is unable to move their fingers to that choice. The system can again offer prompting and encouragement (e.g., “I know that you know the answer.”) to promote the desired movement.

Sensory Regulation - Both tablet and AR systems can engage multiple learning channels (e.g., video and audio) and support control over voice characteristics (R3). However, AR provides additional features to better support our requirements. For example, realistic virtual avatars of a trusted educator or caregiver can be embedded within the learning environment for regulation (R3). Furthermore, virtual objects, e.g., virtual letterboards, can move with a user and be configured to remain always within their visual performance field. This helps accommodate learners who move constantly to regulate themselves (R3).

Furthermore, hand, head, and eye gaze tracking can be leveraged to reduce anxiety. For example, input objects, e.g., virtual objects depicting choices, can be repositioned and reoriented if the tracking data indicates user frustration in reaching these objects. Finally, data from a wearable sensor, e.g., a FitBit, can be correlated with frame level information pertaining to the learning session. Since heart rate and heart rate variability can be used as proxies for stress [16], this would allow educators to modify the learning environment e.g., select a different lesson or offer a sensory break, in a data-driven manner to reduce anxiety.

Tracking progress - Data from the AR device can be used to objectively assess the progress of a learner (R4). For example, hand tracking data can be used to characterize the distance between the correct and actual location of pointing. This information can in turn be used to adapt the motor skills mode employed by the system. We select AR as opposed to VR for several reasons. AR is less immersive than VR. Since autistic people can experience sensory overload, AR is a more appropriate choice. Furthermore, AR affords the potential to interact with actual people, e.g., caregivers and therapists, to foster better regulation, as discussed previously.

A preliminary prototype of the HoloType system has been implemented for Microsoft’s HoloLens 2 Head Mounted Device (HMD). Figure 1 shows the components in the system and their interactions. The system consists of a backend Web-based lesson generator module that helps an educator automate the customization of a standard lesson for the beginner, intermediate, and advanced modes described in Section 3. The module uses Natural Language Processing (NLP) models [2, 18, 28] to automatically generate questions pertaining to a lesson’s content as appropriate for a given mode, e.g., 2 choices for the beginner mode. It also automatically selects words from the lesson and external contextual words related to the lesson for spelling exercises. The auto generated beginner, intermediate, and advanced versions of the standard lesson can then be manually refined by the educator, if desired.

The frontend module executes on the HoloLens 2. It allows an educator to select from a catalog of pre-generated lessons in the repository. It also allows them to select motor skill levels. Optionally, data from a wrist-worn sensor, e.g., FitBit, can be integrated by the system for further analysis.

HoloType delivers the lesson in an interactive manner as discussed previously. The learner is engaged using multiple learning channels. Specifically, the audio lesson is closed captioned. Furthermore, spelling animations are used to complement the virtual

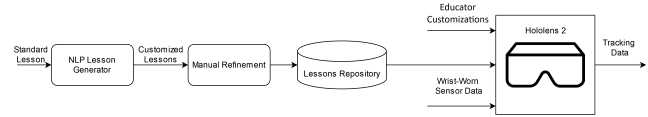
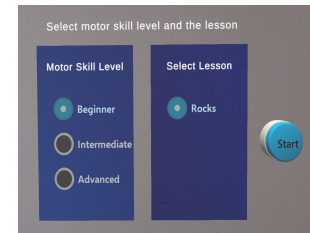
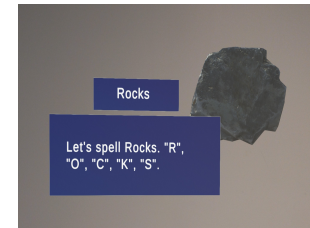


Figure 1: System Components. Standard lessons can be auto-customized using the NLP module for motor skills. The lessons can then be refined manually and delivered within HoloLens 2. Data from a session can be used to customize the learning environment.

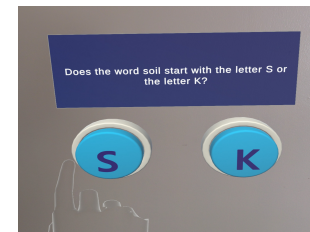
educator led spelling initiatives. All input devices, e.g., virtual letter boards, can be configured to move with the user. We have also developed an analytics dashboard that correlates wrist-worn sensor data from the learner with detailed frame-level events in the lesson. Figure 2 provides sample screenshots from the prototype. We



(a) Motor skills configuration and lesson selection by therapist.



(b) Spelling words related to the lesson visually complemented by voice.



(c) Prompting a beginner user in a spelling exercise.

Figure 2: Lesson on rocks delivered by HoloType

are currently in the process implementing the remaining requirements and sharing the prototype more widely with the nonspeaking

community. We will also commence human subject studies to understand how our prototypes can be refined to facilitate a better user experience.

5 CONCLUSIONS AND FUTURE WORK

This paper presents a prototype system designed to improve communication outcomes for nonspeaking autistic people. HoloType uses AR technology to deliver interactive educational content that simultaneously allows users to improve their pointing skills. Improved pointing skills are critical for developing expressive written communication. In contrast to existing solutions, we incorporate the lived experience of autistic users and their educators in our prototype. Specifically, our design requirements were driven by two sources of information. First, we considered publicly recorded testimonials of nonspeaking autistic people who have learned to type. Second, we obtained insights from a group of experts who have taught non-speaking autistic learners to communicate through typing. These requirements reveal systematic techniques to improve motor skills and strategies to engage learners by addressing their sensory and regulation needs. Future work will conduct an extensive and rigorous evaluation of the ideas embodied in the prototype with nonspeaking autistic users.

REFERENCES

- [1] Douglas Biklen. 2005. *Autism and the myth of the person alone*. New York University Press.
- [2] Steven Bird, Ewan Klein, and Edward Loper. 2009. *Natural language processing with Python: analyzing text with the natural language toolkit*. "O'Reilly Media, Inc."
- [3] ANDY BONDY and LORI FROST. 2001. The Picture Exchange Communication System. *Sage Publications Thousand Oaks • London • New Delhi* 25, 5 (2001), 725–744.
- [4] Amanda Brignell, Karen V Chenausky, Huan Song, Jianwei Zhu, Chen Suo, and Angela T Morgan. 2018. Communication interventions for autism spectrum disorder in minimally verbal children. *Cochrane Database of Systematic Reviews* 11 (2018).
- [5] Miao-En Chien, Cyun-Meng Jheng, Ni-Miao Lin, Hsien-Hui Tang, Paul Tael, Wen-Sheng Tseng, and Mike Y. Chen. 2015. iCAN: A tablet-based pedagogical system for improving communication skills of children with autism. *International Journal of Human-Computer Studies* 73 (2015), 79–90. <https://doi.org/10.1016/j.ijhcs.2014.06.001>
- [6] James A Coan and David A Sbarra. 2015. Social baseline theory: The social regulation of risk and effort. *Current opinion in psychology* 1 (2015), 87–91.
- [7] Charlotte DiStefano, Wendy Shih, Ann Kaiser, Rebecca Landa, and Connie Kasari. 2016. Communication growth in minimally verbal children with ASD: The importance of interaction. *Autism Research* 9, 10 (2016), 1093–1102.
- [8] Kimberly A Fournier, Chris J Hass, Sagar K Naik, Neha Lodha, and James H Cauraugh. 2010. Motor coordination in autism spectrum disorders: a synthesis and meta-analysis. *Journal of autism and developmental disorders* 40, 10 (2010), 1227–1240.
- [9] Morton Ann Gernsbacher. 2004. Language is more than speech: a case study. *The journal of developmental and learning disorders* 8 (2004), 79.
- [10] Naoki Higashida. 2013. *The reason I jump: The inner voice of a thirteen-year-old boy with autism*. Knopf Canada.
- [11] Ido Kedar. 2016. *Ido in Autismland. Climbing Out of Autism's Silent Prison*. Ido Kedar.
- [12] Martha R Leary and Anne M Donnellan. 1995. *Autism: Sensory-movement differences and diversity*. Cambridge Book Review Press.
- [13] Martha R Leary and David A Hill. 1996. Moving on: autism and movement disturbance. *Mental Retardation-Washington* 34, 1 (1996), 39–53.
- [14] Yael Leitner. 2014. The co-occurrence of autism and attention deficit hyperactivity disorder in children—what do we know? *Frontiers in human neuroscience* 8 (2014), 268.
- [15] Patricia Mesa-Gresa, Hermenegildo Gil-Gómez, José-Antonio Lozano-Quilis, and José-Antonio Gil-Gómez. 2018. Effectiveness of virtual reality for children and adolescents with autism spectrum disorder: an evidence-based systematic review. *Sensors* 18, 8 (2018), 2486.
- [16] Benjamin W Nelson, Carissa A Low, Nicholas Jacobson, Patricia Areán, John Torous, and Nicholas B Allen. 2020. Guidelines for wrist-worn consumer wearable assessment of heart rate in biobehavioral research. *NPJ digital medicine* 3, 1 (2020), 1–9.
- [17] Edlyn Vallejo Peña. 2019. *Leaders Around Me: Autobiographies of Autistics who Type, Point, & Spell to Communicate*.
- [18] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. 2019. Exploring the limits of transfer learning with a unified text-to-text transformer. *arXiv preprint arXiv:1910.10683* (2019).
- [19] Caroline E Robertson and Simon Baron-Cohen. 2017. Sensory perception in autism. *Nature Reviews Neuroscience* 18, 11 (2017), 671–684.
- [20] Jodi Robledo, Anne Mary Donnellan, and Karen Strandt-Conroy. 2012. An exploration of sensory and movement differences from the perspective of individuals with autism. *Frontiers in integrative neuroscience* 6 (2012), 107.
- [21] Jacqui Rodgers and A Ofield. 2018. Understanding, recognising and treating co-occurring anxiety in autism. *Current developmental disorders reports* 5, 1 (2018), 58–64.
- [22] Robert Rosenthal and Lenore Jacobson. 1968. Pygmalion in the classroom. *The urban review* 3, 1 (1968), 16–20.
- [23] Dustin E Sarver, Mark D Rapport, Michael J Kofler, Joseph S Raiker, and Lauren M Friedman. 2015. Hyperactivity in attention-deficit/hyperactivity disorder (ADHD): Impairing deficit or compensatory behavior? *Journal of abnormal child psychology* 43, 7 (2015), 1219–1232.
- [24] David James Savarese. 2017. Deej.
- [25] Matt Tincani and Kathryn Devis. 2011. Quantitative synthesis and component analysis of single-participant studies on the picture exchange communication system. *Remedial and Special Education* 32, 6 (2011), 458–470.
- [26] Elizabeth B Torres, Maria Brincker, Robert W Isenhour III, Polina Yanovich, Kimberly A Stigler, John I Nurnberger Jr, Dimitri N Metaxas, and Jorge V José. 2013. Autism: the micro-movement perspective. *Frontiers in integrative neuroscience* 7 (2013), 32.
- [27] Christie Welch, Helene Polatajko, Patty Rigby, and Margaret Fitch. 2019. Autism inside out: lessons from the memoirs of three minimally verbal youths. *Disability and rehabilitation* 41, 19 (2019), 2308–2316.
- [28] Boon Peng Yap, Andrew Koh, and Eng Siong Chng. 2020. Adapting BERT for Word Sense Disambiguation with Gloss Selection Objective and Example Sentences. *arXiv preprint arXiv:2009.11795* (2020).
- [29] Casey J Zampella, Leah AL Wang, Margaret Haley, Anne G Hutchinson, and Ashley de Marchena. 2021. Motor skill differences in autism spectrum disorder: A clinically focused review. *Current psychiatry reports* 23, 10 (2021), 1–11.